

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake

Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Case Study

Code of Conduct:

I Savitha R / 192421368 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SAVITHA R

Annexure A

1. Intelligent Disease Diagnosis Using Bayesian Methods

Conceptual Understanding

A hospital can use a probabilistic diagnosis system based on **Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks (BBN)**. Bayes' theorem updates the probability of a disease when new symptoms are observed.

$$P(\text{Disease} | \text{Symptoms}) = [P(\text{Symptoms} | \text{Disease}) \times P(\text{Disease})] / P(\text{Symptoms})$$

A Naïve Bayes classifier assumes that symptoms are conditionally independent given the disease:

$$P(\text{Disease} | X) \propto P(\text{Disease}) \times \prod_i P(X_i | \text{Disease})$$

Application

The system can use features such as **fever, cough, blood pressure, age, medical history, and laboratory results**. A BBN can represent dependencies between diseases and symptoms.

For example:

Disease $\rightarrow$ Fever

Disease $\rightarrow$ Cough

Disease $\rightarrow$ Blood Test

When new patient evidence is entered, posterior probabilities are calculated and the disease with the highest probability can be suggested.

Analysis

Probability learning estimates disease and symptom probabilities from historical patient data. Uncertainty is handled using prior and posterior probabilities. If symptoms are dependent, a BBN is more suitable than Naïve Bayes because it explicitly represents dependencies.

Real-World Constraints

Medical data may be incomplete, noisy, imbalanced, or limited. Patient privacy and model interpretability are also important. The system should therefore support doctors rather than completely replace clinical judgment.

Conclusion

A Bayesian diagnostic system can provide **probabilistic and explainable disease predictions**, while handling uncertainty and updating predictions when new medical evidence becomes available.

2. Spam Detection Using Naïve Bayes and Bayes Optimal Classifier

Conceptual Understanding

A spam detection system can classify emails as **Spam** or **Not Spam** using Naïve Bayes. Email features can include words, sender information, links, and message frequency.

$$P(\text{Spam} | X) \propto P(\text{Spam}) \times \prod_i P(X_i | \text{Spam})$$

The **Bayes Optimal Classifier** considers predictions from all possible hypotheses and selects the class with the highest posterior probability.

Application

The system can be designed as:

Email → Preprocessing → Feature Extraction → Naïve Bayes → Spam/Not Spam

Text preprocessing includes tokenization, removal of unnecessary words, and feature extraction such as word frequencies.

MLE vs MDL

Maximum Likelihood Estimation (MLE) selects parameters that maximize the likelihood of the observed training data.

$$\hat{\theta}_{MLE} = \arg \max_{\theta} P(D | \theta)$$

Minimum Description Length (MDL) selects a model that provides a short description of both the model and the data.

MDL generally prefers simpler models, while MLE focuses more strongly on fitting the observed data.

Analysis

With a small sample size, complex hypothesis spaces can cause overfitting. More training data usually improves classification reliability. Increasing model complexity may improve training accuracy but can reduce generalization.

Conclusion

Naïve Bayes is fast and effective for spam detection. **MLE provides good data fitting, while MDL helps control model complexity**, producing a balance between accuracy and generalization.

3. Customer Segmentation Using EM Algorithm

Conceptual Understanding

The **Expectation-Maximization (EM) algorithm** is used when data contains hidden or unobserved variables. It can identify customer groups and estimate their underlying probability distributions.

For example, customers may belong to hidden groups such as:

Customer Data → EM Algorithm → Budget / Regular / Premium Customers

Application

Customer features can include:

- Purchase frequency
- Average spending
- Product categories

- Recency of purchase

The EM algorithm works iteratively:

E-Step: Estimate the probability that each customer belongs to each cluster.

M-Step: Update the parameters of each cluster using these probabilities.

This process continues until the likelihood or parameter changes become sufficiently small.

Analysis

EM can converge to a **local maximum**, so different initializations may produce different results. Poor initialization can lead to slow or incorrect convergence. Multiple initializations can be used to select a better solution.

Probability learning helps estimate the hidden cluster membership and underlying distributions.

Application and Limitations

The resulting clusters can support **targeted advertisements, personalized offers, and customer recommendations**.

However, EM is sensitive to initialization, outliers, model assumptions, and the selected number of clusters.

Conclusion

EM is useful for discovering hidden customer patterns in incomplete data, but **careful initialization, convergence monitoring, and validation** are necessary for reliable segmentation.

4. Medical Image Classification Using Candidate Elimination

Conceptual Understanding

Candidate Elimination is a concept learning algorithm that searches a **hypothesis space** using labelled examples. It maintains:

- **S boundary:** Most specific hypotheses
- **G boundary:** Most general hypotheses

The algorithm updates these boundaries using positive and negative examples.

Application

For medical images, extracted features could include:

- Texture
- Shape
- Intensity
- Edge characteristics

The training examples are labelled as:

Image Features → Candidate Elimination → Normal / Abnormal

Positive examples make the hypothesis more general, while negative examples make it more specific.

Finite vs Infinite Hypothesis Space

Finite hypothesis space:

- Contains a limited number of possible hypotheses.
- Easier to search and analyze.
- Requires fewer computational resources.

Infinite hypothesis space:

- Contains potentially unlimited hypotheses.
- More flexible but computationally difficult.

- Requires more data for reliable generalization.

Sample Complexity and Mistake Bound

A larger hypothesis space generally requires more training examples to identify the correct concept. The **mistake bound model** estimates the maximum number of prediction mistakes before the learner identifies a consistent hypothesis.

Analysis

After training, the system should be tested using unseen medical images. Cross-validation can help evaluate generalization. In practice, raw medical images are usually too complex for basic Candidate Elimination, so meaningful feature extraction is required.

Conclusion

Candidate Elimination demonstrates hypothesis-space search effectively, but its performance depends strongly on **hypothesis-space size, training examples, feature representation, and noise**.

5. Investment Prediction Using Bayesian Inference and Gibbs Algorithm

Conceptual Understanding

A financial decision system can use **Bayesian inference** to predict investment outcomes under uncertainty. It combines prior knowledge with observed market data.

$$P(\theta | D) \propto P(D | \theta) P(\theta)$$

where $P(\theta)$ is the prior and $P(\theta | D)$ is the posterior distribution.

Application

The system can use:

- Stock prices
- Market trends
- Interest rates
- Economic indicators
- Trading volume

The model initially assigns prior probabilities to possible market conditions. As new market data arrives, Bayesian inference updates these beliefs.

Gibbs Algorithm

The **Gibbs sampling algorithm** can be used to generate samples from a complex posterior distribution. It repeatedly samples each parameter conditional on the current values of the other parameters.

The generated samples can be used to estimate probabilities such as:

$$P(\text{Return} > 0 \mid \text{Market Data})$$

and

$$P(\text{Loss} > \text{Threshold} \mid \text{Market Data})$$

Analysis

Strong prior knowledge can influence predictions, especially when limited data is available. As more data becomes available, the likelihood increasingly influences the posterior.

Posterior probabilities can help estimate financial risk and support decisions such as whether to invest, hold, or reduce investment.

Computational Challenges

Gibbs sampling may require many iterations and can converge slowly when variables are highly correlated. Burn-in, convergence checking, efficient sampling, and appropriate initialization can improve performance.

Conclusion

Bayesian inference with Gibbs sampling provides a flexible approach for **dynamic financial prediction and risk estimation**, because beliefs can be continuously updated as new market information becomes available.